

Austin Li

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PROJECTS

Field Oriented Control OneWheel

- Built a brushless PMSM motor with a custom distributed winding stator and 6 pole rotor to achieve a sinusoidal back-EMF profile
- **Defined requirements** for a high power electronic speed controller and sourced components to follow a budget constraint and tight timeline
- Designed schematic sheets, PCB layouts, and layer stackup in **Altium** for an ESC with FOC capabilities and an on-board 6-axis IMU for lean-activated control
- Designed a **168 MHz** field oriented control (FOC) algorithm utilizing I2C and SPI protocols for low-side current sensing and 14 bit IMU communication, and SVPWM current control to run on an STM32F405 MCU

Autonomous Pool-Playing Robot

- Created a custom VEX robot with **C++** firmware to successfully detect a pool ball using a laser sensor and strike it into the pocket of a pool table
- Designed a compact servo actuated scissor mechanism in **SolidWorks** to predictably launch a pool ball in a straight path across the table
- **Debugged** inaccuracies in ball-location program and added rolling averages and outlier filters resulting in an improvement to **100% success rate**

Cycloidal Gearbox Robotic Joints

- Designed a 1:14 cycloidal reducer for use in backdrivable joints on a robotic arm
- Validated prototypes with FDM printing and utilized brass bushings resulting in reduced friction by **30%**
- Used **FMEA** during design phases to add reinforcement to force-loaded joints

EXPERIENCE

Waterloo Rocketry

Oct 2025 - Present

Active Controls Canards R&D

Waterloo, ON

- Developed the teams first worm gear-driven canards prototype with magnetic encoders to fit a **57% smaller** footprint and allow for 3-axis control during flight
- Launched with a **successful roll control** program from my subteam to an altitude of 63,000 ft at Launch Canada 2026, setting a **new world record height** for amateur liquid-engine rockets

CiF Lab Solutions

May 2026 - Aug 2026

Automated Machine Specialist

Toronto, ON

- Led the production team of 4 to operate an automated color application assembly line comprised of 19 machine units; oversaw quality inspection on 500+ parts/day
- Operated a 3-axis CNC and adjusted G-code scripts for improved efficiency and accuracy

EDUCATION

University of Waterloo | 4.0 GPA

BASc Mechatronics Engineering

Waterloo, ON

- **Coursework:** Data Structures, Algorithms, Mechanics of Deformable Solids, System Design, Electronics

SKILLS & INTERESTS

Design Tools : Solidworks, Fusion 360, Onshape, Altium, KiCad

Software : C++, Python, Java, Arduino, Simulink

Interests : F1, Pole vault, Golf